

Lecture 02

Bill Perry

Review of Lecture 1

Covered

- Inductive vs deductive reasoning
- Formulating research questions
- Accuracy vs precision
- Data types and classifications
- Setting up R projects
- Installing and loading libraries
- Reading files into R
- Creating basic graphs

Lecture 2: Project Design & Data Visualization

The objectives:

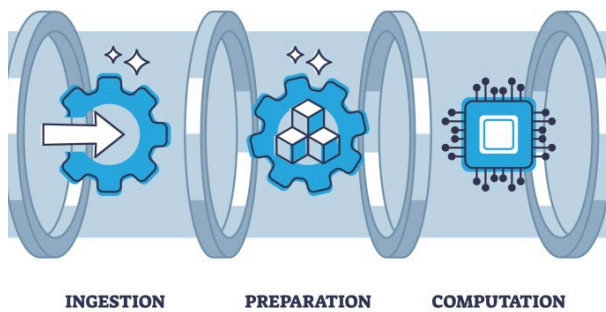
1. Design a well-organized project
2. Implement good naming conventions
 - Controlled vocabulary
 - Including units in names
3. Create and use metadata effectively
4. Build tidy, well-structured spreadsheets
5. Understand data repositories
6. Create effective visualizations with ggplot2



Project Design: Step 1

- Data: the raw material of science
- Wide variety of formats, sizes, complexity
- Data management and curation often under emphasized
- Good data management: owe it to our funding agencies, colleagues, supervisors, and study systems

DATA PIPELINE

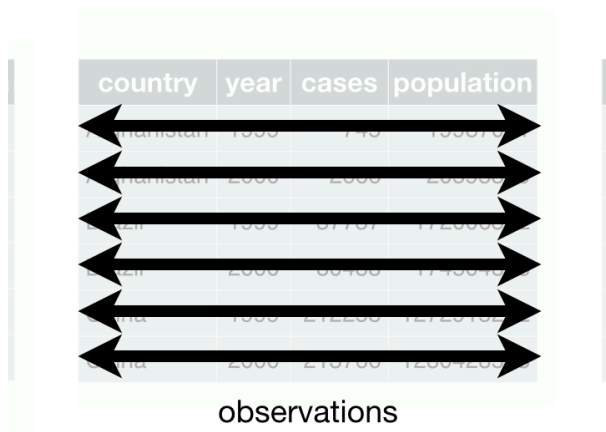


Lecture 2: Project Design: Step 1

1. **Determine data types** you'll collect
2. **Establish controlled vocabulary**
 - Example: `do_mgl` for dissolved oxygen in mg/L
 - Example: `drp_uql` for dissolved reactive phosphorus in $\mu\text{g/L}$
3. **Plan your data flow** from collection to analysis
4. **Organize your project structure** (folders, files)
5. **Enter data promptly** after collection
6. **Save in multiple formats** (Excel and CSV)
7. **Ensure tidy data principles** from the start

Note

See Hadley Wickham's Tidy Data principles for best practices



Project Design: Step 2

Create a **Metadata Sheet** that includes:

- Variable descriptions
- Units of measurement
- Collection methods

- Instrument details
- Dates and locations
- Any other relevant contextual information

country	year	cases	population
iran	1999	143	150074
iran	2000	200	200074
iran	1999	8740	1720074
iran	2000	88400	1740074
iran	1999	212200	1272074
iran	2000	210700	120072074

observations

Practice Exercise 1: Pine Data Organization

💡 Practice Exercise 1: Pine Data Organization

Let's examine our pine needle data: - What naming conventions did you choose? - How did you organize the data? - How can you verify data formats (numeric vs categorical)? - What's your plan for organizing outputs and figures?

```
# Code to read and examine data
library(tidyverse)
library(patchwork)
library(flextable)

pine_df <- read_csv("data/pine_needles.csv")
pine_df
```

```
# A tibble: 48 × 6
  date      group      n_s  wind  tree_no length_mm
<chr>   <chr>    <chr> <chr>   <dbl>     <dbl>
1 3/20/25 cephalopods n    lee      1         20
2 3/20/25 cephalopods n    lee      1         21
3 3/20/25 cephalopods n    lee      1         23
4 3/20/25 cephalopods n    lee      1         25
5 3/20/25 cephalopods n    lee      1         21
6 3/20/25 cephalopods n    lee      1         16
7 3/20/25 cephalopods s    wind      1         15
8 3/20/25 cephalopods s    wind      1         16
9 3/20/25 cephalopods s    wind      1         14
10 3/20/25 cephalopods s    wind      1         17
# i 38 more rows
```

Lecture 2: Data Management: Step 3

Storage and Backup Strategy:

1. Store raw data and metadata securely
 - Save in both Excel and CSV formats
 - Consider write-protecting raw data files
2. Implement the 3-2-1 backup rule:
 - 3 total copies of data
 - 2 different storage media
 - 1 offsite location (cloud storage)
3. Establish a regular backup schedule



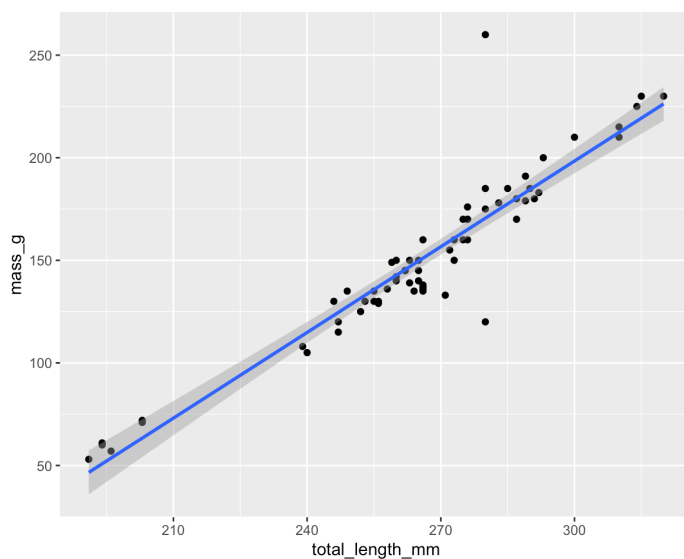
Lecture 2: Data Management: Step 4

Initial Data Inspection:

1. Examine data in the Environment tab
2. Run `summary()` and `glimpse()` functions
3. Create exploratory visualizations
4. Check for outliers, errors, and missing data

```
# Specify how to handle missing values during import
pine_df <- read_csv("data/pine_needles.csv",
                    na = c("", "NA", "N/A", "missing", "null"))

# Get a quick summary
summary(pine_df)
```



Practice Exercise 2: Try plotting a histogram

💡 Practice Exercise 2: Try plotting a histogram of your data

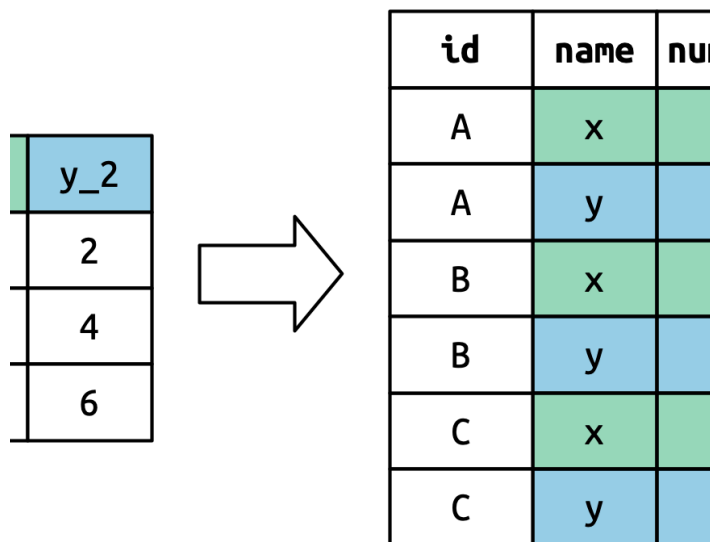
Create a histogram of pine needle lengths to check the distribution:

```
# Write your code here to make a plot
# How do you examine the data - what are the ways you think and lets try it!
```

Lecture 2: Data Management: Step 5

Data Cleaning:

1. Correct errors and inconsistencies
2. Replace missing values with proper NA codes
3. Document all changes made to raw data
4. Save a clean, master version (consider making read-only)
5. Keep notes on data cleaning procedures



Lecture 2: Data Management: Step 6

Analysis and Visualization Workflow:

1. Create exploratory visualizations
2. Summarize and transform data as needed
3. Document all analysis steps
4. Save outputs systematically

A good way to organize script files is number them in the order they get run.

ewater_import_clean.R

ewater_initial_raw_means_figures.R

ewater_means_mixed_anova.R

ewater_anova_mixed_graphs with a fi

ewater_anova_mixed_graphs.R

ewater_graphs_functions.R

ewater_figs_jeq_formatting.R

mary statistics.R

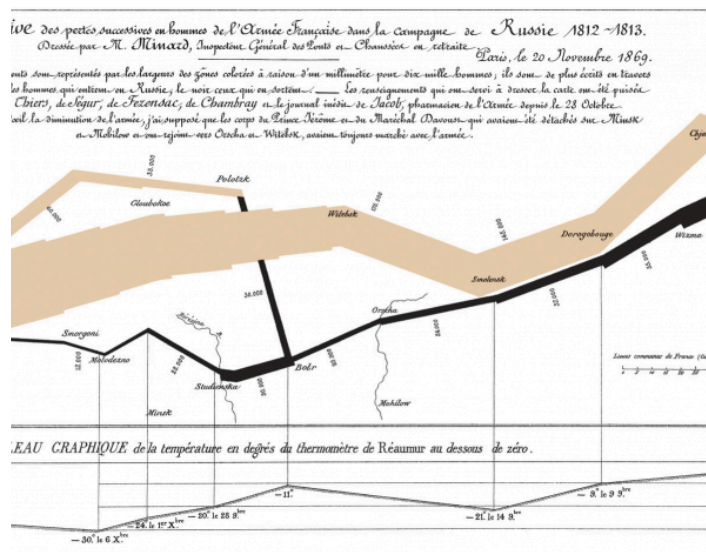
Lecture 2: Effective Data Visualization

Why make plots?

Get in a group and discuss

- What is the purpose of a data visualization?
- What elements are essential in an effective plot?
- What characteristics define a “good” plot?
- What common mistakes make plots ineffective?

Napoleon’s Disastrous Invasion of Russia Detailed in an 1869 Data Visualization: It’s Been Called “the Best Statistical Graphic Ever Drawn”



Lecture 2: Tables vs. Visualizations

How readable are tables?

We will get to what these number mean and how to make them in the next lecture.

- Tables
 - ▶ are they useful in a presentation?

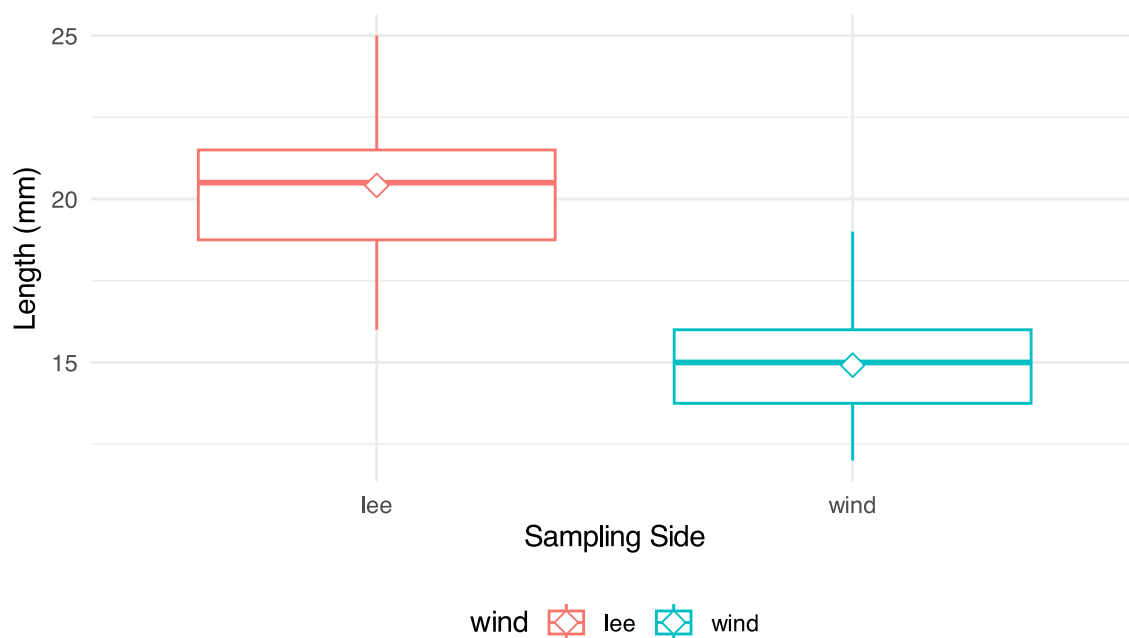
wind	n	mean_mm	sd_mm	min_mms	max_mm
lee	24	20	2.45	16	25
wind	24	15	1.91	12	19

Lecture 2: Displaying data

- how does a table compare to a plot?
- Does this help?
- What is this plot?
 - if you don't explain does the audience know?

Pine Needle Length on wind and lee sides of a tree

Same data as the table



Lecture 2: Principles of Effective Graphics

According to Tufte (2001), good scientific graphics:

1. **Show the data** without distortion
2. **Maximize data-ink ratio** (minimize non-data elements)
3. **Make large datasets coherent** and understandable
4. **Encourage comparison** between elements
5. **Reveal multiple layers** of information
6. **Serve a clear purpose** in telling your story
7. **Integrate with statistical methods** appropriately

```
# A tibble: 4 × 7
  group      mean_mm sd_mm      n se_mm conf_low conf_high
<chr>      <dbl> <dbl> <int> <dbl>   <dbl>   <dbl>
1 cephalopods    18  3.86    12  1.11    15.5    20.5
2 crayfish       18  3.86    12  1.11    15.5    20.5
```

3	salmon	16.3	3.94	12	1.14	13.8	18.8
4	snail	18.3	2.27	12	0.655	16.9	19.8

Lecture 2: Creating Effective Graphics

According to Tufte (2001), good scientific graphics:

- To implement these principles:
 - Focus on the data, not decorative elements
 - Ensure proportional representation of numbers
 - Provide clear and informative labels
 - Remove unnecessary elements (“chart junk”)
 - Revise and refine visualizations iteratively

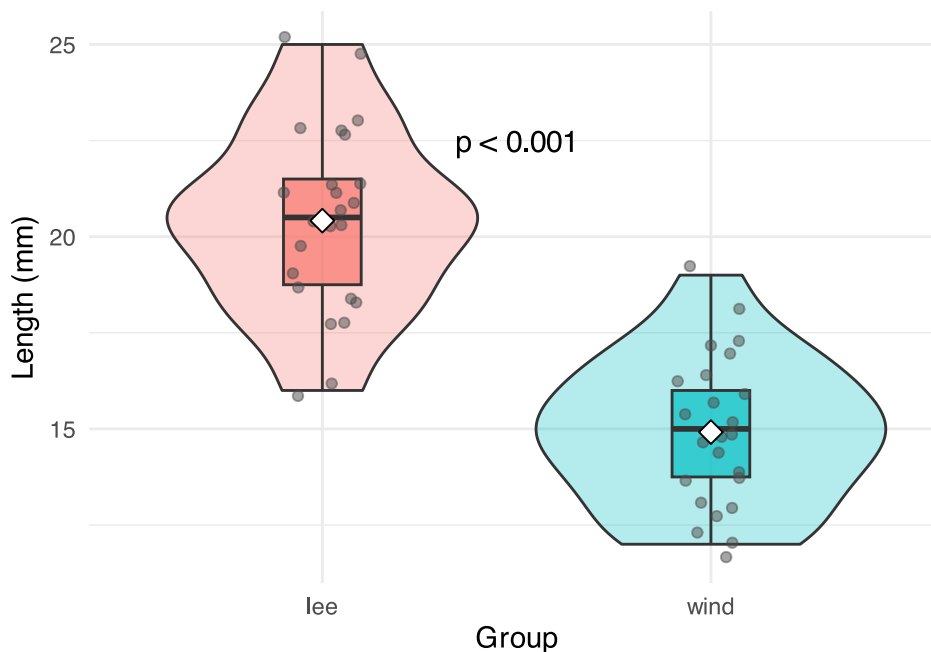
Lecture 2: Displaying data - Good Graphics

To make good graphics:

- Above all, focus on data
- Do not distort data
- Graphical representation of numbers → directly proportional to numbers
- Strive for clarity through labeling
- Maximize data-ink ratio
 - Remove non-data ink
 - Reduce redundant data ink
- Revise and redraw

Pine Needle Length by Group

Multiple layers: raw data, distribution, central tendency, CI



Lecture 2: Displaying data - Poor Example

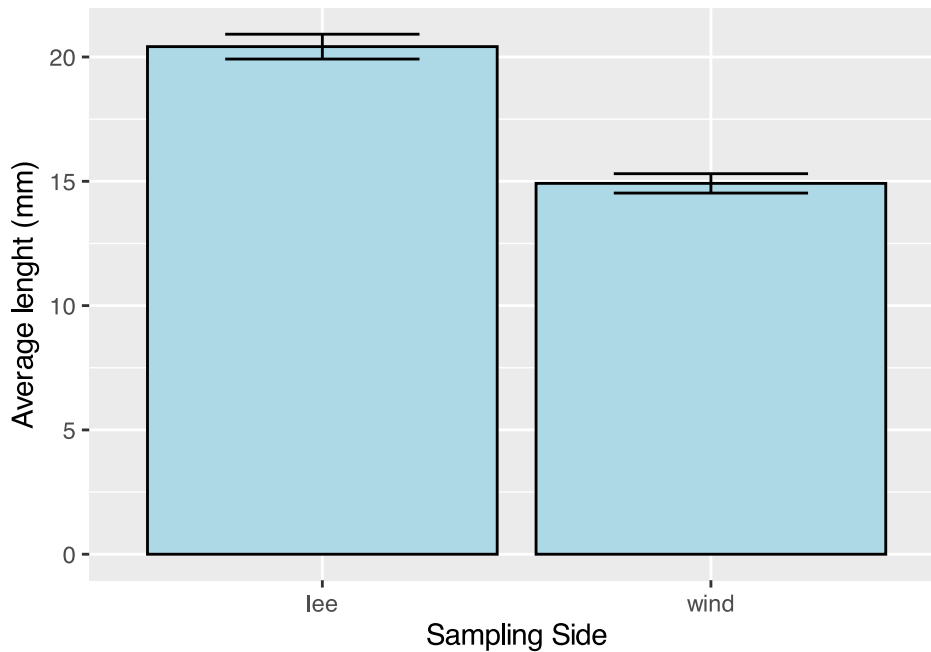
What do you think?

Does this -

- Focus on data
- Distort data
- Is it directly proportional to numbers
- Is labeling clear
- Maximize data-ink ratio
 - Remove non-data ink
 - Reduce redundant data ink
- Revise and redraw

Average needle length

This plot has a low data-ink ratio



Lecture 2: Displaying data - Better Example

What do you think?

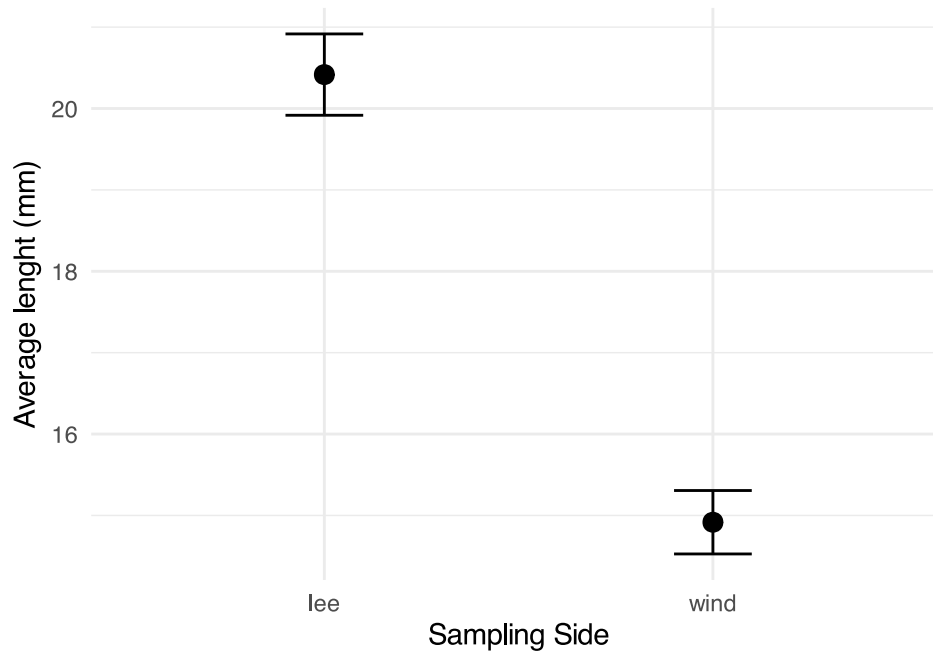
Does this -

- Focus on data
- Distort data
- Is it directly proportional to numbers
- Is labeling clear
- Maximize data-ink ratio
 - Remove non-data ink
 - Reduce redundant data ink
- Revise and redraw

What is one of the most common plots you make all the time?

Average needle length

This plot has a low data-ink ratio



Lecture 2: Displaying data - Common Problems

Common Visualization Problems

1. Data distortion:

- Non-zero baselines on bar charts
- 3D effects that skew perspective
- Inappropriate scales

2. Excessive “chart junk”:

- Too many gridlines
- Unnecessary decorative elements
- Redundant information

3. Poor color choices:

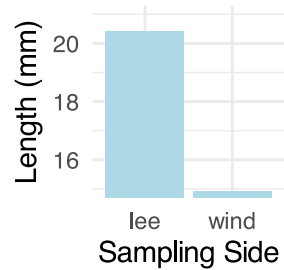
- Too many colors
- Non-colorblind-friendly palettes
- Colors that don't print well in grayscale

4. Misleading representations:

- Pie charts with too many categories
- Dual y-axes with different scales
- Truncated axes without clear indication

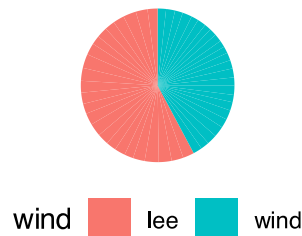
Misleading: Truncated Y-axis

Exaggerates differences



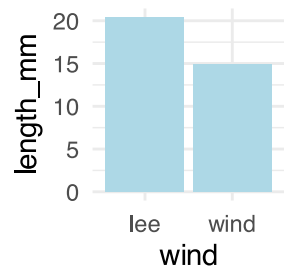
Misleading: Pie Chart

Hard to compare values



Misleading: Dual Axes

Suggests false correlation



Practice Exercise 3: Basic Plots with Pine Data

💡 Practice Exercise 3: Lets try some plots with pine data first

Lets try to make some basic plots

```
# Write your code here to make a dot plot or X y plot
# How do you examine the data - what are the ways you think and lets try it!
# what is missing - hwo do you tell the effect of wind?
```

Practice Exercise 4: Colors, Shapes, and Fills

💡 Practice Exercise 4: OK we are closer but what about colors or shape or fills

Lets try to make some more basic plots

This is free time - we will free code this....

Below are some examples of code you will need for the future

```
# Write your code here to make a dot plot or X y plot
# How do you examine the data - what are the ways you think and lets try it!
# what is missing - hwo do you tell the effect of wind?
```

Lecture 2: Introduction to the Grammar of Graphics - ggPLOT

We will learn the anatomy of a GGplot is layers

- ggplot2 uses a **layered grammar of graphics** approach:
 1. **Data:** The dataset you're visualizing
 2. **Aesthetics:** Mapping variables to visual properties
 3. **Geometries:** The visual elements representing data
 4. **Facets:** Splitting visualization into subplots
 5. **Statistics:** Statistical transformations of the data
 6. **Coordinates:** The space in which data is plotted
 7. **Themes:** Overall visual style of the plotWe have aesthetics

Lecture 2: Building a ggplot Visualization

Key Components:

1. **Aesthetics (aes)** map variables to visual properties:
 - x and y positions
 - color, fill, shape, size, alpha
 - group, linetype
2. **Geometries (geom_*)** determine how data is displayed:
 - `geom_point()`: Scatter plots
 - `geom_line()`: Line graphs
 - `geom_boxplot()`: Box-and-whisker plots
 - `geom_violin()`: Violin plots
 - `geom_histogram()`: Histograms
 - `geom_bar()`: Bar charts
3. **Position adjustments** control how elements are arranged:
 - `position_dodge()`: Side-by-side elements
 - `position_jitter()`: Add random noise to points
 - `position_stack()`: Stack elements on top of each other
4. **Labels and annotations** provide context:
 - `labs()`: Title, subtitle, caption, axis labels
 - `annotate()`: Add text, shapes, etc.

Lecture 2: Fine-tuning your visualizations

1. Colors, fills, and shapes:

```
scale_color_manual(
  values = c("wind" = "darkblue", "lee" = "darkred"),
  labels = c("wind" = "Windward", "lee" = "Leeward")
)
```

2. Coordinate systems:

```
coord_cartesian(ylim = c(10, 30)) # Zoom in without dropping data
```

3. Themes:

```
theme_minimal() +
theme(
  axis.title = element_text(size = 14),
  legend.position = "bottom"
)
```

4. Combining plots with patchwork:

```
plot1 + plot2 + plot_layout(ncol = 2)
```

Practice Exercise 5: Publication-Quality Plot

💡 Practice Exercise 4: Creating a Publication-Quality Plot

Create a fully customized plot that would be suitable for publication:

```
# Create a publication-quality plot
pine_df %>%
  ggplot(aes(x = wind, y = length_mm, fill = wind)) +
  geom_violin(alpha = 0.4) +
  geom_boxplot(width = 0.2, alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.1, alpha = 0.5, color = "gray30", size = 2) +
  stat_summary(fun = mean, geom = "point", shape = 23, size = 3, fill = "white") +
  labs(
    title = "Pine Needle Length Varies with Wind Exposure",
    subtitle = "Needles on the leeward side tend to be longer",
    x = "Tree Side",
    y = "Needle Length (mm)",
    caption = "Data collected Spring 2023") +
  scale_fill_manual(
    values = c("wind" = "#1b9e77", "lee" = "#d95f02"),
    labels = c("wind" = "Windward", "lee" = "Leeward")) +
  theme_minimal() +
  theme(
    plot.title = element_text(face = "bold", size = 16),
    plot.subtitle = element_text(size = 12, color = "gray30"),
    axis.title = element_text(face = "bold"),
    legend.title = element_blank(),
    legend.position = "bottom")
```

Key Takeaways

1. **Plan your data management** from the beginning
 - Consistent naming conventions
 - Good organization
 - Regular backups
2. **Make your data tidy** from the start
 - One observation per row
 - One variable per column
 - One value per cell
3. **Create effective visualizations** by:
 - Focusing on data, not decoration
 - Using appropriate plot types
 - Following good design principles
 - Customizing for clear communication
4. **Master the grammar of graphics** to:
 - Build plots layer by layer
 - Communicate patterns clearly
 - Tell compelling stories with data

Next Steps

- Practice creating different types of plots
- Learn to combine multiple plots effectively
- Explore statistical transformations in `ggplot2`
- Develop a consistent visualization style